Preferences Toward Risk, Uncertainty, and Safety in Financial Decisions (Students vs. IT Professionals)

Team 8:
Parth Dhodapkar 2024111009
Joshua Koilpillai 2024101128

1. Problem Statement

Standard economic models assume people make financial decisions with consistent, rational risk tolerance based only on net worth. In reality, choices depend on everyday context: specifically, how steady someone's income is and what bills they have to pay.

We study this by comparing university students and IT professionals across four practical groups (technically, it's just two groups, University Students and IT professionals. The following 4 classes arise with higher sample sizes):

1. **Unstable, Low Responsibility:** Students supported by family. They get small allowances or stipends, have no dependents or debt, and rarely touch financial markets.
2. **Stable, Low Responsibility:** Early career IT workers with good disposable income, living cheaply with minimal commitments.
3. **Stable, High Responsibility:** IT professionals earning steady salaries, but paying rent, education or home loans, and supporting family members.
4. **Unstable, High Responsibility:** Working students funding their own tuition and daily living expenses.

Participants evaluate three types of options:

1. **Risk:** Choices with known probabilities (e.g., a 50/50 coin flip).
2. **Uncertainty:** Ambiguous choices where odds are unknown.
3. **Safety:** Options with zero chance of losing money.

We want to answer two questions: Does a steady salary make IT workers take more risks, or do their bills make them play it safe? And do students avoid unknown odds due to lack of experience (and money), or take bigger gambles because they have little to lose?

2. Methodology and Analysis

We will run a free, self-administered survey via Google Forms or Microsoft Forms (or any other similar service) targeting university students and working IT professionals.

2.1 Calculating S and R

Participants first answer intake questions about their cash flow and liabilities. We convert these into two scores from 0.0 to 1.0:

- **Income Stability Score (S):** Average of monthly income predictability, earnings dependability over the last six months, and liquid savings measured in months of living expenses.
- **Financial Responsibility Score (R) :** Average of dependent count, share of monthly money spent on non-negotiable living costs, and active debt obligations.

2.2 Decision Scenarios and Scoring

Participants work through practical scenarios: allocating a windfall, picking between a fixed deposit and an equity fund, choosing a cash-to-stock ratio, and reacting to a 40% market crash.

We track risk appetite and ambiguity tolerance as two separate scores so they do not distort each other:

- **Risk vs. Safety:**
  Questions with known odds feed into a **Risk Appetite Score (0 to 100)**:
  - **Safety Bucket (0 to 49):** Prefers guaranteed payouts, protects capital, and holds mostly cash.

    - **Risk Bucket (50 to 100):** Takes calculated risks when expected payouts justify volatility.

- **Ambiguity Tolerance Index:** Scenarios with hidden odds (such as picking a mystery token box over a 50/50 box, or backing an unproven startup over a known franchise) yield an **Ambiguity Tolerance Score (0 to 100)**. This lets us test whether salary stability actually reduces fear of the unknown, or if ambiguity aversion affects both groups equally.

2.3 Visualizations and Tests

We plot the data directly to observe natural patterns rather than forcing a mathematical curve:

1. **Scatter Plot:** Plot each person at (S, R), colored by their bucket (Safety vs. Risk), to see where boundaries or clusters form.
2. **Cohort Centroids:** Plot mean (S, R) coordinates for students and IT professionals to show the baseline difference between the two groups.
3. **Ambiguity Comparison:** Compare mean Ambiguity Tolerance across cohorts and buckets to observe the behaviour of professionals and students under uncertain conditions.
4. **Crash Reaction**: Compare how students versus IT professionals react to a 40% market crash (selling, holding, or buying more) using a chi-square test to see if steady earners handle panic better.